TEAM ZOLLERQU

Portfolio Selection with QAOA



Which stocks to buy? One qubit per asset, sampled on the IonQ Forte 1 noise model.

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This effort is supported by Qollab & IonQ

PROBLEM

Which assets should a fund buy?

- Banks and funds build portfolios every day
- They want low risk and a good return
- Risk comes from assets that fall together

Every extra asset doubles the choices

AAPL Apple	**GLD** gold	**XOM** oil	**JPM** bank
TLT US bonds	**JNJ** health	**PG** consumer	**XLU** utilities
VNQ real estate	**FCX** copper	**MSFT** Microsoft	**SLV** silver

4,096
portfolios, 12 assets

536,870,912
portfolios, 29 assets

Data: weekly prices of 12 US stocks and funds, 2006-2014 (Kaggle Huge Stock Market Dataset, CC0)

FORMULATION

One qubit per asset, one Ising energy

Score to minimise: risk minus return

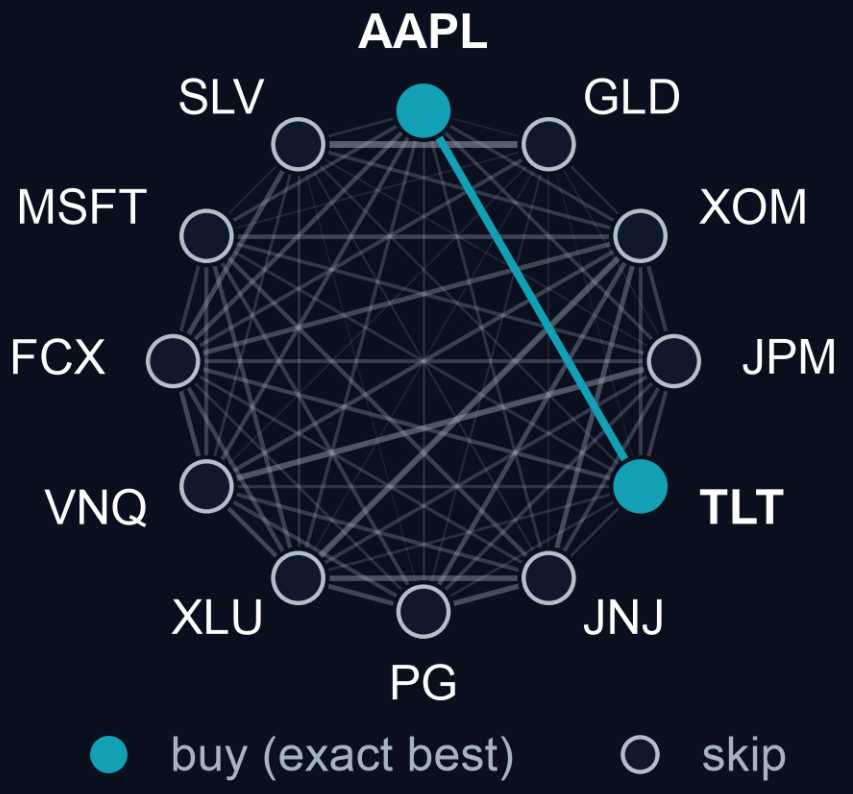
$$J(x) = \underbrace{\frac{x^{\mathrm{T}} C\, x}{n^2\, V_{\mathrm{ref}}}}_{\text{risk}} - \eta \underbrace{\frac{\mu^{\mathrm{T}} x}{n\, \sigma_{\mathrm{ref}}}}_{\text{return}}$$

The same score as an Ising energy

$$E(s) = \sum_{i<j} J_{ij}\, s_i\, s_j + \sum_i h_i\, s_i + c$$

1 qubit = 1 asset: |0> buy, |1> skip

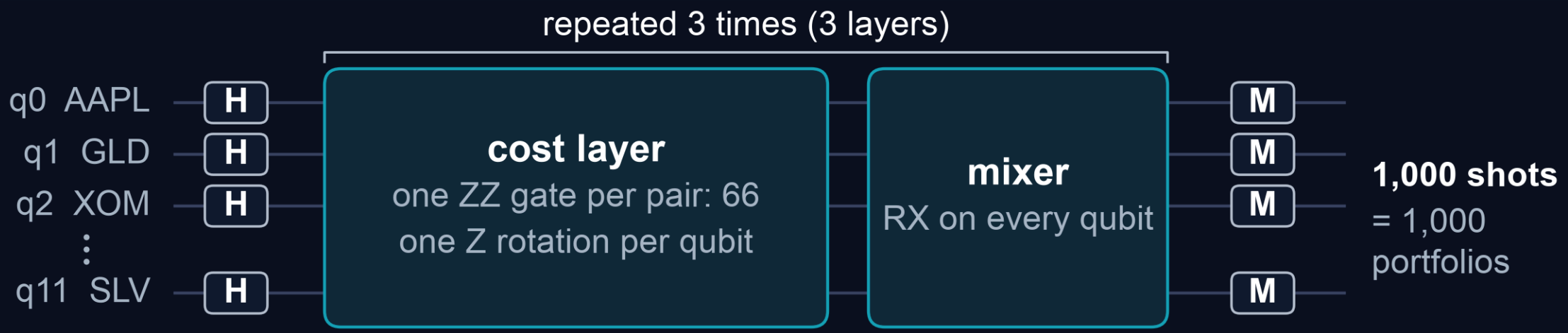
12 qubits · 66 couplings · 4,096 states



Exact best portfolio found by checking all 4,096 portfolios (classical); risk and return from 2006-2014 data

QUANTUM METHOD

QAOA circuits, trained offline, one IonQ job



COMPARED

8 circuit types

58 circuits, trained offline on a classical simulator

BEST AT 12 QUBITS

CVaR-QAOA

3 layers, 198 two-qubit gates, sent to IonQ as one job

AT 16, 20, 29 QUBITS

Linear-ramp QAOA

Angles reused from 12 qubits, no new training

12 qubits: 58 circuits + a random-guess control, noiseless simulation · 16-29 qubits: server simulation

WHAT WE TRIED

8 circuit types, 5 settings that mattered

58 circuits tested

6 kinds of QAOA

- Linear-ramp QAOA
- Free-angle QAOA, 5 optimisers
- CVaR-QAOA (α = 0.1, 0.25)
- Multi-angle QAOA
- Counterdiabatic QAOA
- XY-mixer QAOA (keeps balance)

2 imaginary-time methods

- VarQITE on the QAOA angles
- varQITE ansatz (IonQ + ORNL)

Settings we changed (noiseless)

Mixer sign (12 assets)
as written → opposite sign: 0.004 % → 1.8 % per shot

Objective (12 assets)
mean energy → CVaR 0.1: 1.8 % → 5.8 % per shot

Layers (29 assets)
3 → 16: 6.6 % → 90 % of runs (noiseless)

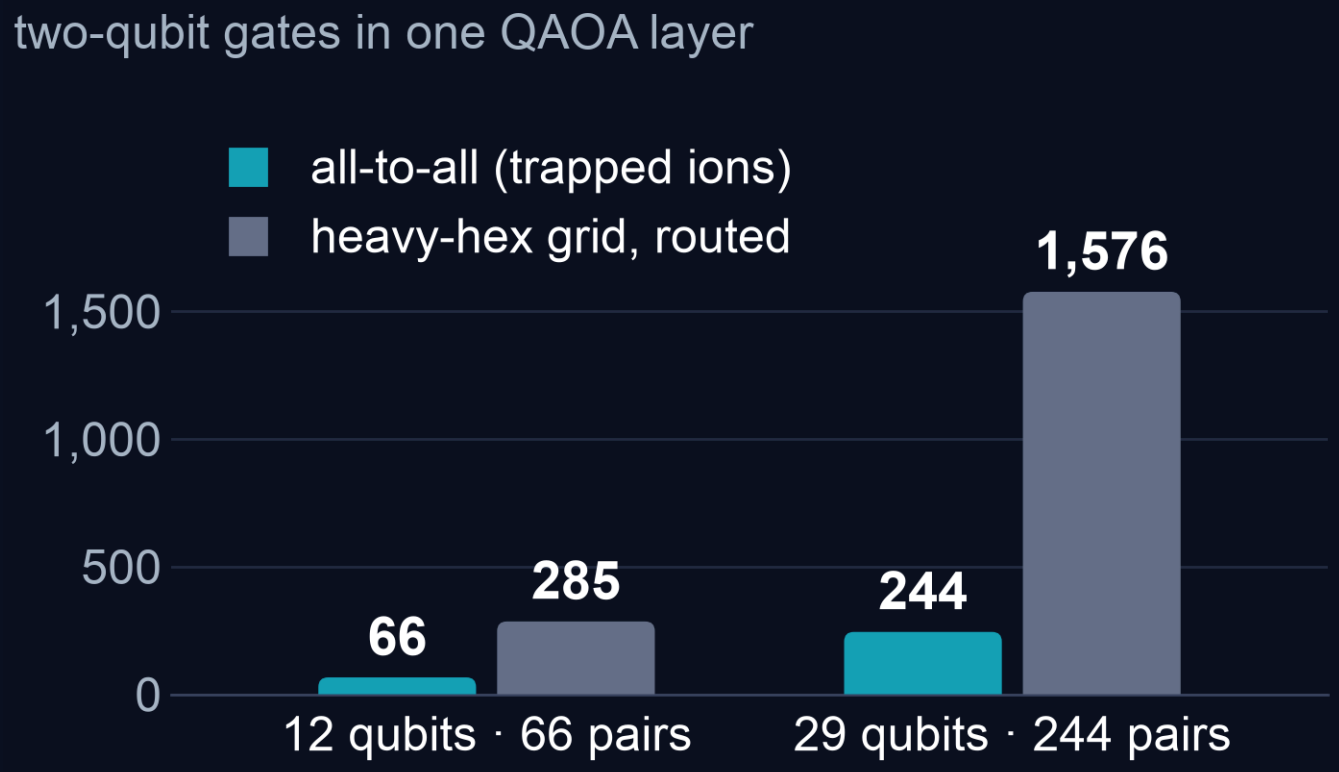
Ramp range (29 assets)
up to 1.5 → 3.0: 45 % → 90 % of runs at 16 layers

Angle choice (29 assets, 3 layers)
by energy → by CVaR: 9.9 % → 17.9 % of runs

12 assets: noiseless simulation, 10 Oct 2026 · 29 assets: noiseless server simulation, 11 Oct 2026
per shot = chance that one shot is the best portfolio · runs = 1,000-shot runs that contain it

WHY IONQ

All-to-all links: 4.3× fewer two-qubit gates



4.3×

fewer two-qubit gates
at 12 qubits (66 vs 285)

6.5×

fewer at 29 qubits (244 vs 1,576)

66 pairs = 66 gates, no SWAPs

Gate counts from transpiling only (Qiskit 2.5.2, level 1); heavy-hex = median of 5 routing seeds on a generic heavy-hex map. No circuit was run on a heavy-hex device.

MAIN RESULT

IonQ noise model: 29× random guessing

IonQ Forte 1 noise model

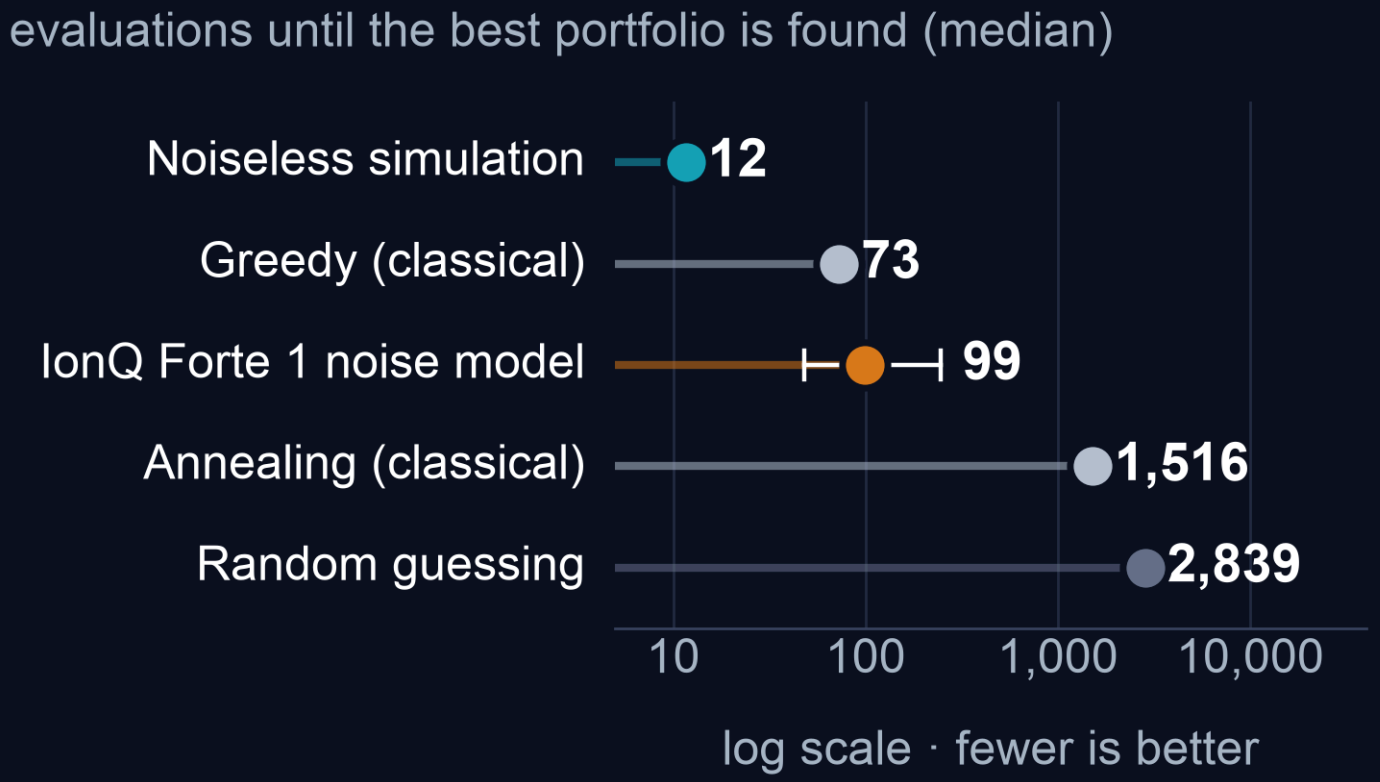
7 / 1,000

shots are the exact best
of 4,096 portfolios

29× random guessing

58 / 1,000 noiseless

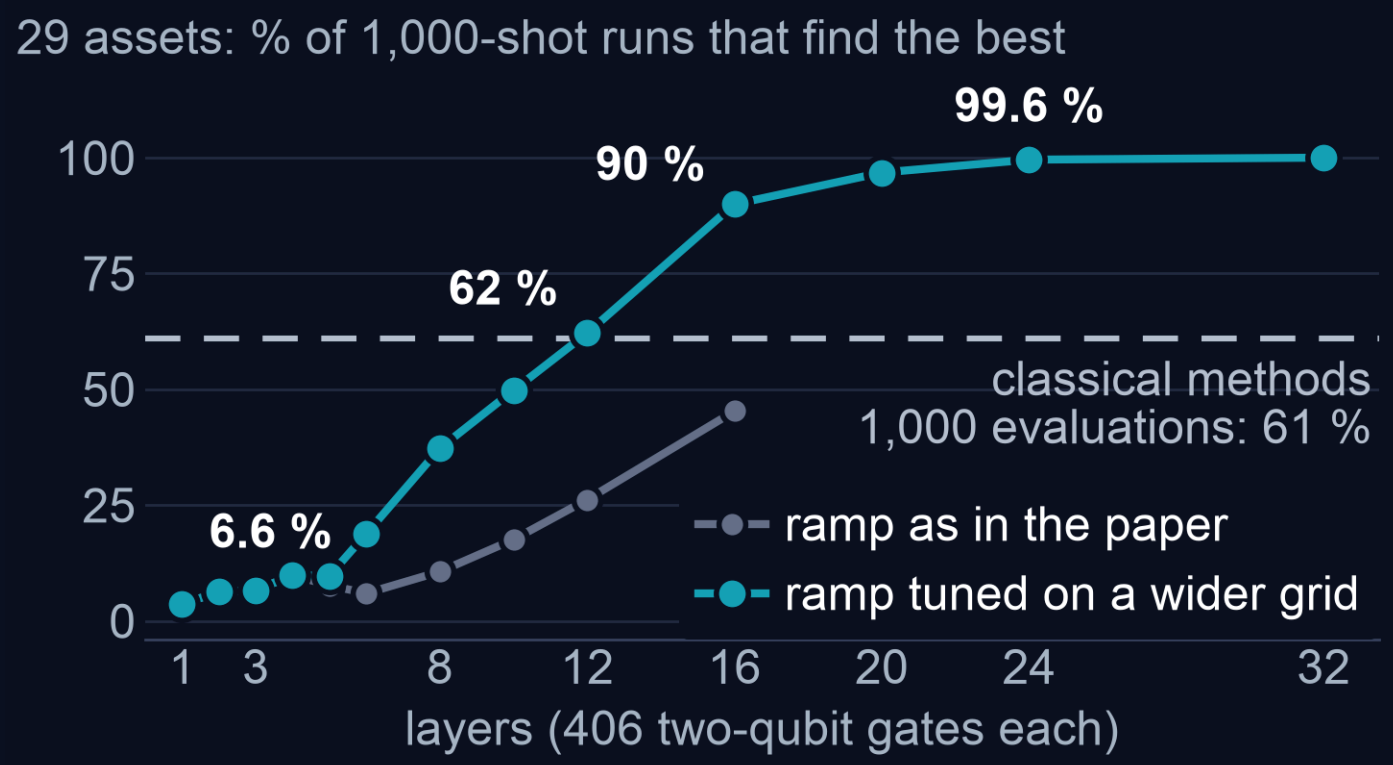
95 % interval: 3-14 per 1,000



IonQ Forte 1 noise model (Qollab): 1 job, 1,000 shots, 11 Oct 2026 · noiseless simulation
Classical: median of 50 runs. Sampler medians come from the hit rate per shot. Training not counted.

SCALING AND HONEST COMPARISON

29 qubits: more layers help, only without noise



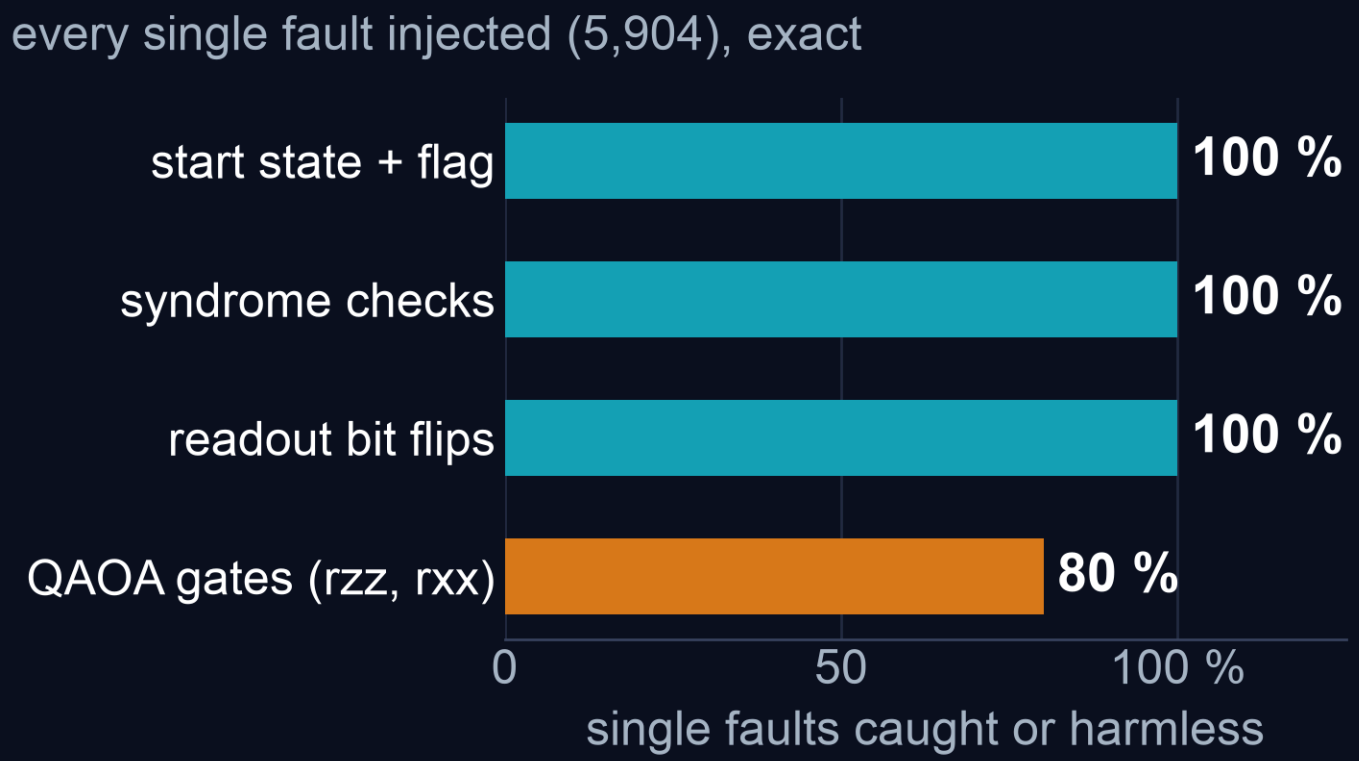
Read with care

- Noiseless simulation only
- 16 layers = 6,496 two-qubit gates; our 198-gate IonQ run lost a factor 8 to noise
- Classical: 100 % with 10,000 evaluations
- A shot is not the same work as a classical evaluation

Server simulation of all 536,870,912 states, 11 Oct 2026; ramp tuned at 12 assets, no tuning at 29
Classical: greedy descent and annealing, 200 runs, the same 1,000 evaluations

ERROR DETECTION

Iceberg code: verified, but it costs gates



17 qubits

12 logical on 14, + 1 flag
+ 2 helpers per check

+58 %

two-qubit gates at 3 layers
(313 vs 198)

- IonQ simulator via Qollab: every check passed, results match the noiseless ones

Faults: exact simulation of every single fault (1, 2 layers) · Qollab IonQ simulator: 12 jobs, 11 Oct 2026
Detection, not correction: shots that fail a check are discarded

DEMO, LIMITS, NEXT STEPS

Try it on Qollab: one click runs it all

Live demo on Qollab

1. Open the project:
   the code is in main.py
2. Run → Select QPU →
   IonQ Forte 1 (29q) noise model
3. Each job returns counts,
   a table and plots

Honest limits

- A noisy simulator: one job of 1,000 shots
- Noise cuts the hits from 58 to 7 per 1,000
- Classical heuristics match or beat it

Next steps

- Iceberg on the Forte 1 noise model
- Shorter circuits: 66 instead of 198 gates
- Find how many layers the noise allows

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Thank you for your attention

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